

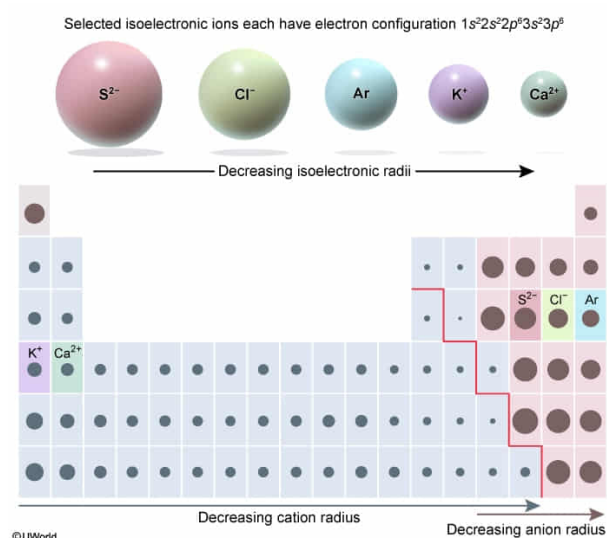
If a channel of a particular cell preferentially transmits smaller ionic species across the membrane, given the following biologically relevant, isoelectronic ions, the channel is LEAST likely to transmit:

- ✔ ☒ A. S^{2-} . [50%]
- ☐ B. Cl^- . [8%]
- ☐ C. K^+ . [20%]
- ☐ D. Ca^{2+} . [21%]

Correct

50% answered correctly

Explanation:



Ionic radii tend to **decrease** in size **across a period** (row) of the periodic table (left to right) and **increase** moving **down a group** (column). This **trend for metal cations** resets and **repeats for anions** beginning near the division between **metals and nonmetals**, past which anions tend to preferentially form.

Compared to the **neutral atom** of a given element, its cation will be smaller but its anion will be larger. Losing electrons to form a cation causes the remaining electrons to experience a greater **effective nuclear charge Z_{eff}** , pulling the electrons closer to the nucleus. Conversely, gaining electrons to form an anion produces greater electronic repulsion and nuclear shielding (lesser Z_{eff}), which pushes electrons farther from the nucleus.

The ions S^{2-} , Cl^- , K^+ , and Ca^{2+} are **isoelectronic** (have the same number of electrons); however, because the number of protons is different in each ion, the electrons in each ion experience a different Z_{eff} . Therefore, in an isoelectronic series, ionic radii decrease as the atomic number increases. Because sulfur has the lowest

atomic number (fewest number of protons) in this isoelectronic series, Z_{eff} is smallest in S^{2-} , making it the largest ion. Given that the cell channel preferentially transmits smaller ions, S^{2-} is least likely to be transmitted.

(Choice B) Anions in an isoelectronic series are generally larger than neutral atoms and cations in the same series, but Cl^- is not the largest because S^{2-} has one less proton (lesser Z_{eff} to pull the electrons toward the nucleus).

(Choices C and D) Cations are generally smaller than anions and neutral atoms in an isoelectronic series.

Educational objective:

Ionic radii decrease in size across a period and increase down a group on the periodic table. This trend occurs for metal cations and then resets and repeats for anions beginning near the division between metals and nonmetals. The use of an isoelectronic series is helpful in comparing cations, anions, and neutral atoms.

Time spent:0 sec

Subject:

General Chemistry

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System:

Atomic Theory and Chemical Composition